S-Band RHCP Patch Antenna

Simulation Report

Rogers RO6010 Substrate · 2.45 GHz · HFSS 2025 R2.4

1. Document Control

Version	Date	Description
V0	2026-05-13	Initial document format
V1	2026-05-14	Updated with figures and performance
V2	2026-05-31	Updated with Efficiency

2. Simulation Environment

Item	Value	Notes
Simulation Software	**Ansys HFSS 2025 R2.4**	
Solver Type	**Frequency Domain FEM**	*Adaptive mesh refinement*
Frequency Sweep Type	**Interpolating + Discrete**	*Discrete used for far-field AR vs freq*
Adaptive Convergence	**ΔS < 0.02**	
Radiation Boundary	**Radiation (PML equivalent)**	*Air box dimensions: 300*300*200 mm*
Port Type	**Coaxial Wave Port**	*Terminal S-parameter solution* *Copper pin, Teflon_based dieletric* *Inner radius = 0.635 mm* *Outer radius = 2.065 mm*

3. Design Parameters

3.1 Substrate

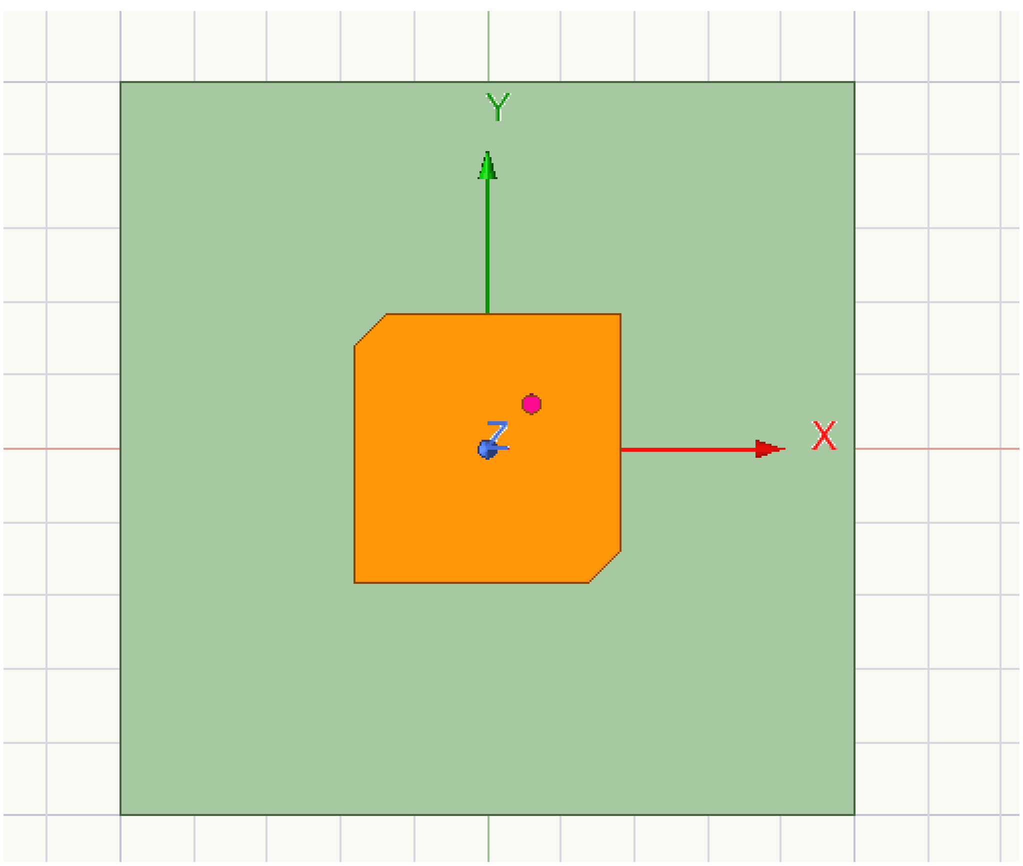
Parameter	Value	Notes
Material	**Rogers RO6010**	*Space-qualified laminate*
Relative Permittivity (εr)	**10.2**	*Datasheet value at 10 GHz*
Loss Tangent (tan δ)	**0.0023**	*At 10 GHz*
Substrate Thickness (h)	**2.54 mm**	*Standard Rogers stock thickness*
Substrate Dimensions	**60 × 60 mm**	*Ground plane / substrate extent*
Copper Cladding Thickness	**17 µm (0.5 oz)**	

3.2 Patch Geometry

Parameter	Value	Notes
Patch Shape	**Square**	*Corner-truncated for CP*
Patch Side Length (L)	**18.32 × 18.32 mm**	*Tuned including probe inductance compensation*
Corner Truncation (c)	**2.2 mm**	*Leg of removed right-isosceles triangle*
Truncated Corners	**Top-Left & Bottom-Right**	*Produces RHCP*
Intact Corners	**Top-Right & Bottom-Left**	

3.3 Feed

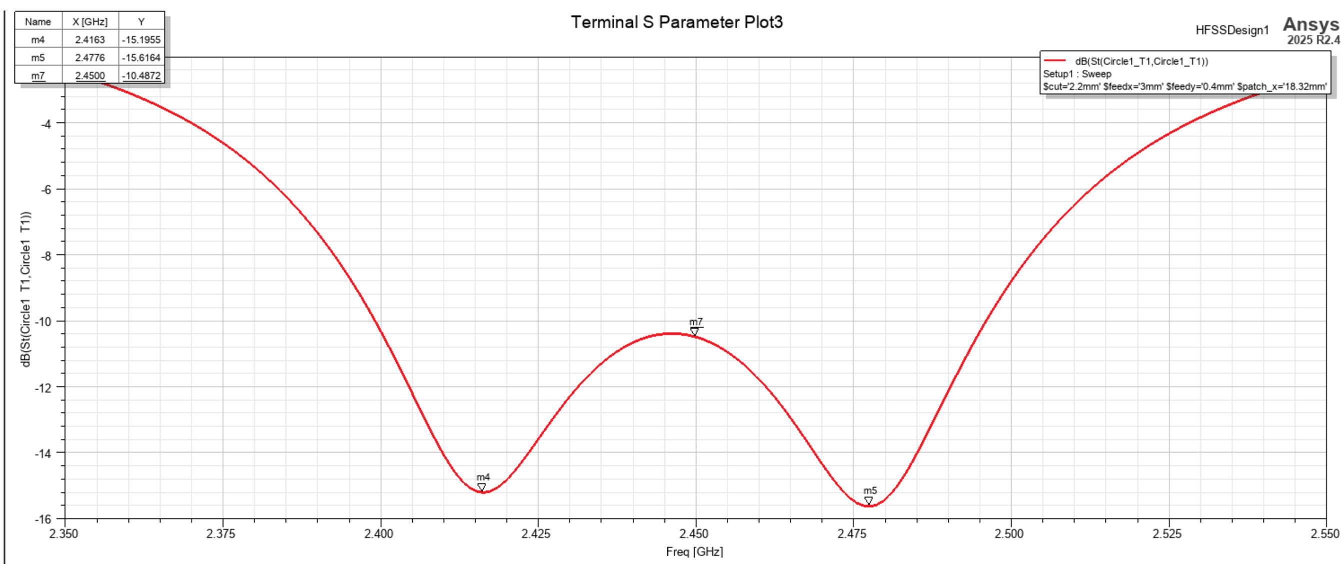
Parameter	Value	Notes
Feed Type	**Coaxial Probe**	*SMA centre pin, 1.27 mm dia.*
Feed X-Offset (feedx)	**3.0 mm**	*From patch centre, +X direction*
Feed Y-Offset (feedy)	**0.4 mm**	*From patch centre, +Y direction*
Feed Impedance	**50 Ω**	
Feed Axis	**Off-diagonal**	*feedx ≠ feedy ensures both modes excited*



4. Simulation Results

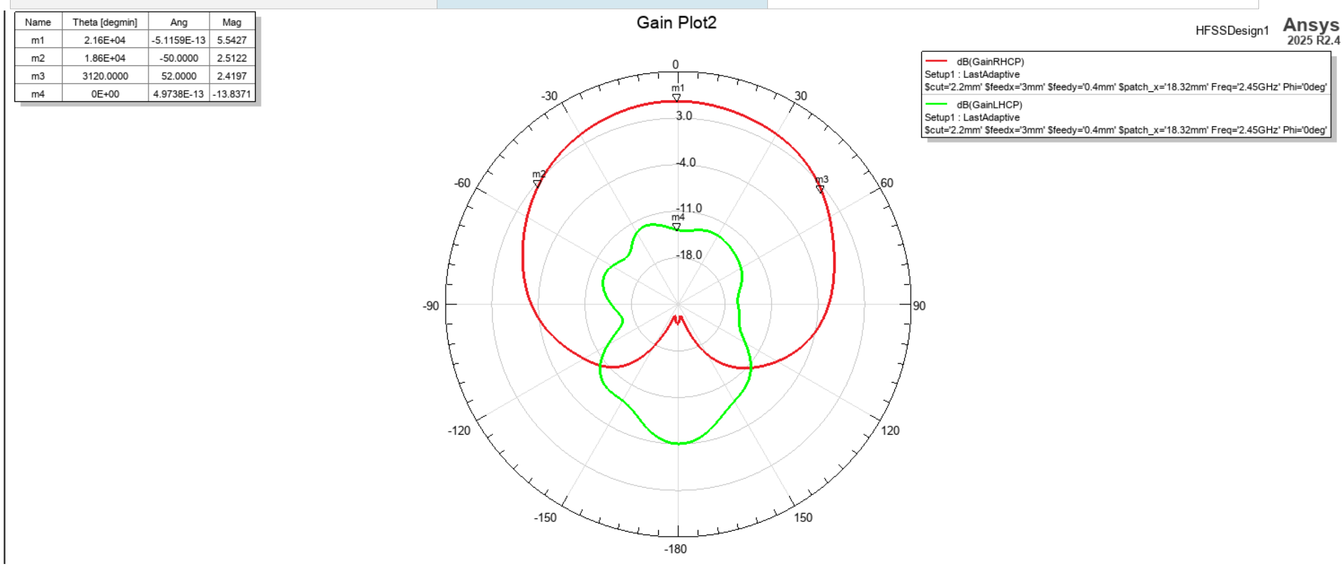
4.1 Impedance — S11

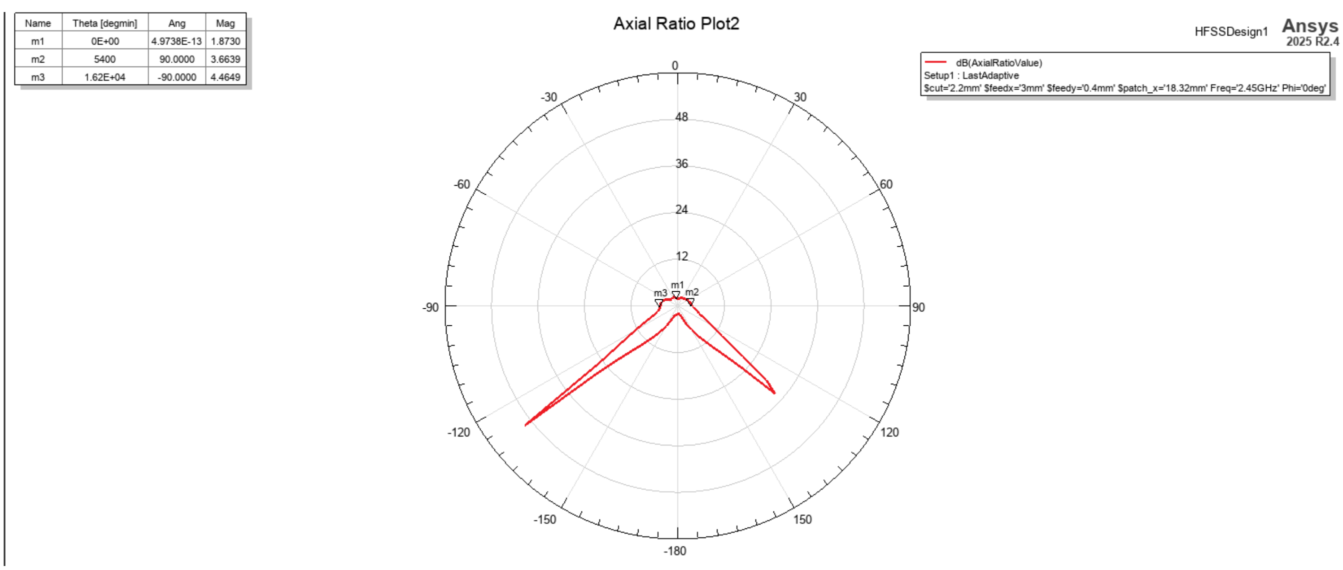
Parameter	Result	Notes
Centre Frequency (f0)	**2.450 GHz**	*Target frequency*
Lower Resonance (f1)	**2.4163 GHz**	*First split mode dip*
Upper Resonance (f2)	**2.4776 GHz**	*Second split mode dip*
Mode Split (f2 − f1)	**61.3 MHz**	*Target ≈ f0 / Q = 60 MHz*
S11 at f1	**−15.20 dB**	
S11 at f2	**−15.62 dB**	
Dip Depth Balance	**0.42 dB**	*< 1 dB*
S11 at f0	**−10.49 dB**	*Midpoint between dips — expected for CP*
Simulated Q Factor	**≈ 41**	*Q = f0 / split = 2450 / 60*

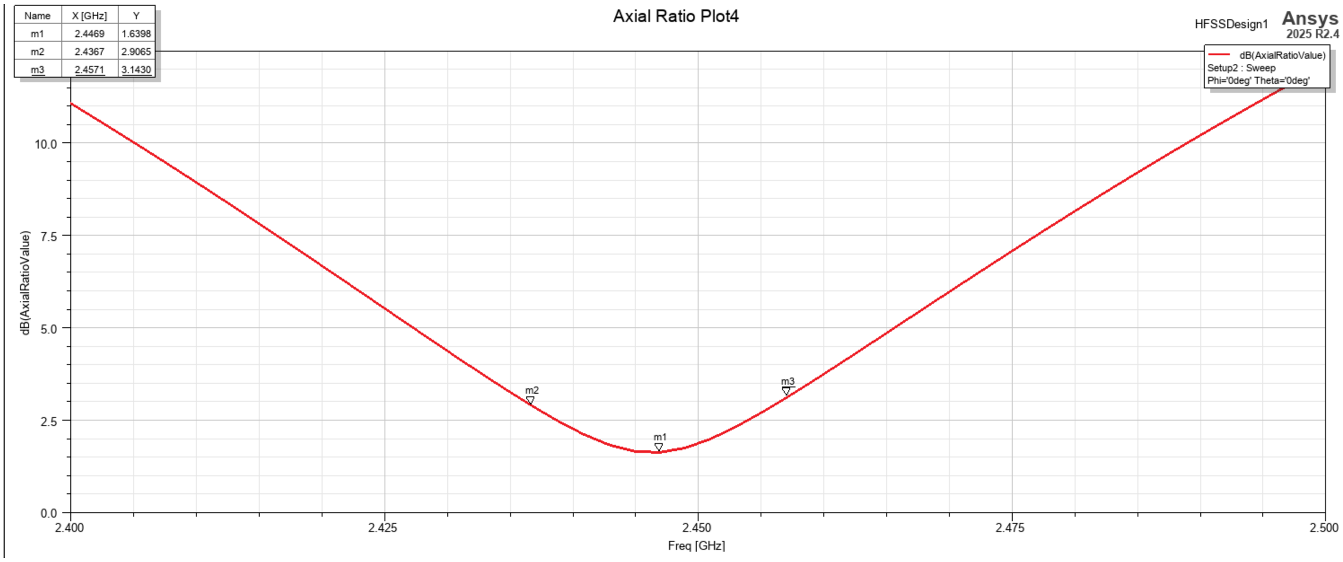


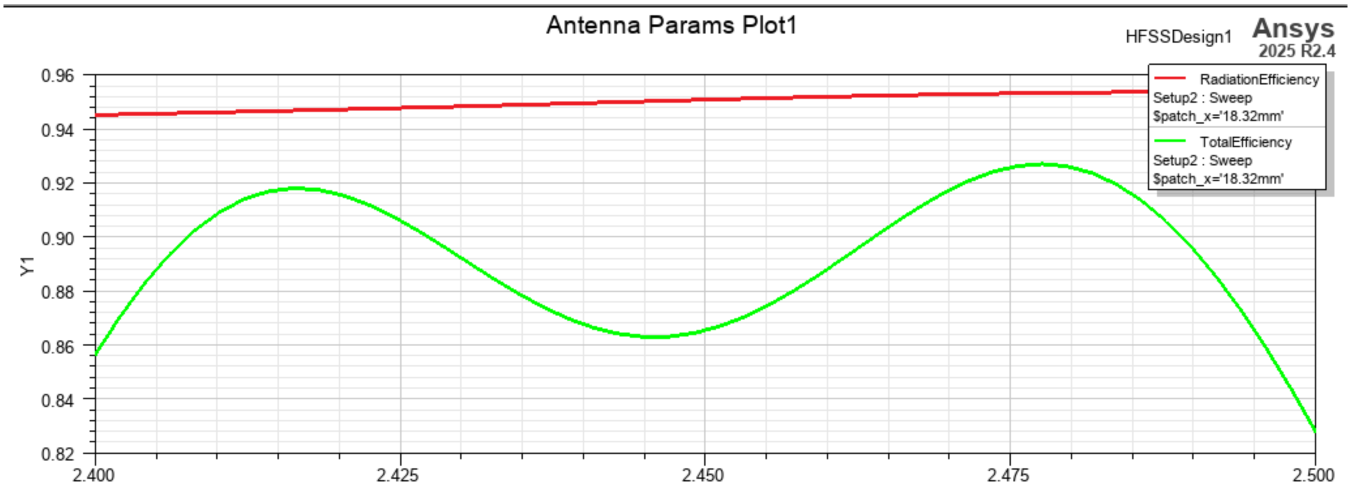
4.2 Radiation Performance

Parameter	Result	Notes
Polarisation Sense	**RHCP**	*Confirmed from GainRHCP >> GainLHCP*
GainRHCP at Boresight	**+5.54 dBi**	*θ = 0°, φ = 0°, f = 2.45 GHz*
GainLHCP at Boresight	**−13.84 dBi**	*Cross-polarisation component*
Cross-Pol Discrimination (XPD)	**19.4 dB**	*GainRHCP − GainLHCP at boresight*
Axial Ratio at Boresight	**1.87 dB**	*< 3 dB threshold*
AR at θ = +90° (horizon)	**3.66 dB**	*Expected degradation off-boresight*
AR at θ = −90° (horizon)	**4.46 dB**	
AR Bandwidth (< 3 dB)	**≈ 20 MHz**	*0.82% fractional BW — sufficient for CubeSat*
Half-Power Beamwidth (RHCP)	**≈ 100°**	*Typical for single patch element*
Radiation Efficiency	**TBD %**	*Extract from HFSS efficiency report*
Total Efficiency	**TBD %**	*Includes mismatch loss*









5. Link Budget Context

Parameter	Value	Notes
Application	**CubeSat S-Band Downlink**	*Qubesat Mission; about 10dB link margin with this antenna*
Target Frequency	**2.45 GHz**	*ISM / amateur satellite band*
Orbit Altitude	**LEO (TBD km)**	*Likely 400 km*
Max Doppler Shift	**±61 kHz**	*At 7.5 km/s orbital velocity*
AR BW vs Doppler Margin	**20 MHz / 0.12 MHz ≈ 167**	*Ample margin*
Required Polarisation	**RHCP**	*Standard ground station convention*

6. Design Summary

Parameter	Result	Status
Centre Frequency	**2.450 GHz**	*PASS; On target*
S11 at f0	**−10.49 dB**	*PASS; < −10 dB*
Mode Split	**61.3 MHz**	*PASS; ≈ f0/Q = 60 MHz*
Dip Balance	**0.42 dB**	*PASS; < 1 dB*
Axial Ratio	**1.87 dB**	*PASS; < 3 dB threshold*
RHCP Gain	**+5.54 dBi**	*PASS; > 4 dBi*
XPD	**19.4 dB**	*PASS; > 15 dB*
AR Bandwidth	**≈ 20 MHz**	*PASS; Sufficient for CubeSat downlink*
Polarisation	**RHCP confirmed**	*PASS; Correct sense*
Overall Status	**PASS — Ready for fabrication review**	

7. Next Steps

The following actions are recommended before fabrication:

#	Action	Owner	Status
1	Extract radiation efficiency and total efficiency from HFSS	Terry	95% Simulated radiation efficiency
~~2~~	~~Measure −10 dB S11 bandwidth from full sweep~~	~~TBD~~	~~Pending~~
3	Fabrication	TBD	Stalled: very expensive